항만 CCTV 해무 분류 ERF 논문 70편 최종 레퍼런스 맵

이 문서는 SCI Q2 급 투고를 염두에 두고, 메인 서사를 유지하면서도 약 70편 수준의 인용 저변을 확보하기 위해 정리한 **curated 70 reference map**이다.

원칙은 다음과 같다.

- 총 70편을 유지하되, 기존 풀에서 논리적으로 약한 6편을 더 강한 6편으로 교체했다.
- 각 항목마다 `들어갈 파트`, `담당 논리`, `논문에 넣을 문장`, `원문 근거 문장`을 함께 기록했다.
- `원문 근거 문장`은 가능한 경우 abstract 또는 Chrome에서 직접 확인한 문장 단위 표현을 사용했고, 불가능한 일부 항목은 fallback 임을 명시했다.
- `기상 기준`, `ICAO 정의`, `KHOA 관측자료`는 논문이 아니라 운영 기준/데이터 출처이므로, 메인 방법론 인용이 아니라 라벨 정의와 운영 맥락 설명용으로만 제한적으로 사용한다.

교체된 6편

- 제외: CBAM, SE-Net, ECA-Net, BAM, Attention Is All You Need, YOLOv3
- 추가: texture bias, BagNet, VISOR-NET, weakly supervised visibility under discrete label distribution, complex-background sea fog detection, CMAda

70편 상세 맵

문제 중요성

No.	기존 번호	논문	넣을 파트	담당 논리	논문에 넣을 문장	원문 근거 문장	근거 출처
1	33	Automatic Sea Fog Detection and Estimation of Visibility on CCTV	Introduction	CCTV 기반 해무/시정 판단의 실용성과 선행 가능성 제시	기존 연구는 CCTV 영상을 이용한 sea fog detection과 visibility estimation의 가능성을 이미 제시하였으며, 이는 항만 감시 영상 기반 시정 분류가 실용적 응용 가치를 가진 문제임을 보여준다.	Choi, Y.; Choe, H.-G.; Choi, J.Y.; Kim, K.T.; Kim, J.-B., and Kim, N.-I., 2018.	OpenAlex abstract
2	34	Sea Fog Dissipation Prediction in Incheon Port and Haeundae Beach Using	Introduction; Related Work	한국 항만 해무 문제의 운영 중요성과 연구 연속성 확보	한국 항만 및 연안 환경에서 sea fog는 실제 운영과 안전에 직접 영향을 주는	Sea fog is a natural phenomenon that reduces the visibility of manned	OpenAlex abstract

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		Machine Learning and Deep Learning			현상이며, 기존의 예측 연구 역시 인천항과 해운대 해역을 대상으로 이러한 중요성을 보고한 바 있다.	vehicles and vessels that rely on the visual interpretation of traffic.	
3	68	Meteorological Grade Standards for Fog and Haze	Introduction; Dataset/Methods	normal/lowvis/seafog 구분의 제도적 시정 기준 제공	본 연구의 시정 등급 구분은 국내 기상 기준에서 사용되는 visibility class definition을 참고하여 설정하였다.	[Open abstract not found] Meteorological Grade Standards for Fog and Haze.	Title-level fallback
4	69	International Civil Aviation Organization Visibility Definitions	Introduction	visibility를 운영 안전과 직접 연결되는 metric으로 정당화	Visibility는 단순한 시각적 품질 변수가 아니라 항행 안전과 직접 연결된 operational indicator이므로, sea fog classification은 실질적인 운용 상태 인식 문제로 볼 수 있다.	[Open abstract not found] International Civil Aviation Organization Visibility Definitions.	Title-level fallback
5	70	KHOA Sea Fog Observation Data	Introduction; Dataset	데이터 출처와 서비스 연계성 설명	본 연구는 KHOA의 sea fog observation framework와 연결된 데이터 맥락 위에서 수행되며, 이는 결과의 현장 적용 가능성을 높여준다.	[Open abstract not found] KHOA Sea Fog Observation Data.	Title-level fallback
6	37	Enhancement of Marine Lantern	Introduction	저시정/해무가 해상 안전과 직접 연결된다는 응용 맥락 보강	저시정과 haze 환경이 해상 표지 식	[Open abstract not found] Enhancement	Title-level fallback

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		Visibility under High Haze Using AI Camera			별에 직접 영향을 준다는 점은, sea fog classification이 단순 학술 문제가 아니라 실제 안전 지원 시스템의 선행 단계가 될 수 있음을 보여준다.	of Marine Lantern Visibility under High Haze Using AI Camera.	
7	53	CNN-Enabled Visibility Enhancement Framework for Vessel Detection under Haze	Introduction; Discussion	visibility degradation이 downstream marine vision task에 미치는 영향 제시	Visibility degradation은 vessel detection과 같은 downstream marine vision task에도 직접 영향을 주므로, sea fog classification의 응용 가치는 단순 상태 분류를 넘어선다.	Maritime images captured under haze environment often have a terrible visual effect, making it easy to overlook important information.	OpenAlex abstract
8	49	Deep learning models for visibility forecasting using climatological data	Introduction	시정 자체가 독립적인 예측/판단 대상이라는 broader visibility literature 연결	Visibility forecasting literature가 보여주듯이 시정 자체는 독립적인 운영 대상이며, sea fog classification 역시 visibility-aware recognition problem으로 이해될 수 있다.	Low visibility conditions affect safety and traffic operations, leading to adverse scenarios that often result in serious accidents.	Chrome/abstract-manual

Shortcut/해석

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9	11	Shortcut learning in deep neural networks	Introduction; Related Work	shortcut learning의 이론 프레임 제시	본 연구는 shortcut learning 관점에 따라, sea fog classifier가 실제 scene-level visibility보다 더 쉬운 local edge-texture proxy에 의존할 수 있다고 가정한다.	Shortcut learning refers to a behaviour where a machine-learning model makes decisions based on 'shortcuts': patterns that allow accurate prediction on training and test data, but do not reflect the underlying phenomenon of interest.	Chrome/abstract-manual
10	new	ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness	Introduction; Related Work; Discussion	CNN의 texture bias를 직접적으로 입증하는 핵심 근거	ImageNet-trained CNN이 shape보다 texture에 더 강한 편향을 보일 수 있다는 결과는, 본 연구에서 관찰된 local edge-texture shortcut 현상을 해석하는 직접적 근거가 된다.	We show that ImageNet-trained CNNs are strongly biased towards recognising textures rather than shapes, which is in stark contrast to human behavioural evidence and reveals fundamentally different classification strategies.	Chrome/abstract-manual
11	new	Approximating CNNs with Bag-of-local-Features models works surprisingly	Introduction; Related Work; Discussion	spatial ordering 없이 local patch evidence 만으로도 높은 성능이 가능함을 제시	Local bag-of-features style model도 높은 분류 성능을 달성할 수 있다는 결과는, sea fog	Our model, a simple variant of the ResNet-50 architecture called BagNet,	Chrome/abstract-manual

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		well on ImageNet			classification 에서도 좋은 accuracy가 곧 scene-level visibility reasoning을 의미하지 않을 수 있음을 시사한다.	classifies an image based on the occurrences of small local image features without taking into account their spatial ordering.	
12	12	On Measuring Localization of Shortcuts in Deep Networks	Related Work; Discussion	shortcut의 공간적 localization을 추적할 필요성 제시	Shortcut을 공간적으로 측정하고 추적하는 관점은, 본 연구의 Grad-CAM 기반 shortcut localization 분석을 정당화한다.	Shortcuts, spurious rules that perform well during training but fail to generalize, present a major challenge to the reliability of deep networks.	Chrome/abstract-manual
13	13	Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization	Methods; Results	ERF 조정 전후 주의 영역 비교의 핵심 시각화 도구	우리는 Grad-CAM을 사용해 ERF 조정 전후 모델의 주의 영역이 local edge에서 scene-level visibility structure로 이동하는지를 시각적으로 확인하였다.	We propose a technique for producing visual explanations for decisions from a large class of CNN-based models, making them more transparent.	Chrome/abstract-manual
14	14	Learning Deep Features for Discriminative Localization	Methods	discriminative localization 계보 보강	Our visualization analysis follows the broader CAM-style tradition of examining which image regions	In this work, we revisit the global average pooling layer proposed in [13], and shed light on how it explicitly enables the	OpenAlex abstract

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					support class decisions.	convolutional neural network (CNN) to have remarkable localization ability despite being trained on imagelevel labels.	
15	15	Grad-CAM++: Generalized Gradient-Based Visual Explanations for Deep Convolutional Networks	Methods; Appendix	필요시 더 세밀한 hotspot localization 대안 제시	보다 세밀한 localization이 필요한 경우 Grad-CAM++는 local hotspot이 구조적 주의 패턴인지 확인하는 대안적 분석 도구가 될 수 있다.	Over the last decade, Convolutional Neural Network (CNN) models have been highly successful in solving complex vision based problems.	OpenAlex abstract
16	16	Axiomatic Attribution for Deep Networks	Discussion	다른 attribution 기법으로 해석 강건성 검증 가능성 제시	향후 본 연구의 shortcut 해석은 integrated gradients와 같은 attribution 기법으로 교차 검증될 수 있다.	We study the problem of attributing the prediction of a deep network to its input features, a problem previously studied by several other works.	OpenAlex abstract
17	new	Visualizing and Understanding Convolutional Networks	Methods; Appendix	perturbation/ablation 기반 해석 tradition 정당화	Grad-CAM과 함께 perturbation-style inspection을 사용하는 것은, 어떤 이미지 영역이 모델 결정에 가장 크게 기여하는지를 살피는 고전적	We introduce a novel visualization technique that gives insight into the function of intermediate feature layers and the	Chrome/abstract-manual

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					CNN visualization tradition과 맞닿아 있다.	operation of the classifier.	

ERF 이론

No.	기존번호	논문	넣을 파트	담당 논리	논문에 넣을 문장	원문 근거 문장	근거 출처
18	1	Understanding the Effective Receptive Field in Deep Convolutional Neural Networks	Introduction; Related Work	actual ERF가 theoretical RF보다 작고 중심 집중적이라는 핵심 이론 근거	Actual effective receptive field는 theoretical receptive field보다 훨씬 작고 중심부에 집중될 수 있으며, 이는 small-kernel backbone이 장면 전체보다 국소 cue에 더 민감할 수 있음을 시사한다.	We study characteristics of receptive fields of units in deep convolutional networks.	OpenAlex abstract
19	2	Scaling Up Your Kernels to 31×31: Revisiting Large Kernel Design in CNNs	Introduction; Related Work; Discussion	large kernel과 broader structural context 활용의 직접적 modern 근거	Large-kernel CNN design literature는 broader spatial context를 활용하는 것이 구조적 단서 이용에 유리할 수 있음을 보여주며, 이는 본 연구의 ERF redesign 동기를 직접 뒷받침한다.	We revisit large kernel design in modern convolutional neural networks (CNNs).	OpenAlex abstract
20	3	A ConvNet for the 2020s	Related Work; Methods	7×7 depthwise kernel이 modern ConvNet 설계의 일부라는 근거	ConvNeXt는 relatively large depthwise kernels가 이미 modern ConvNet design의 중요한 일부임을 보여주며, ERF-oriented backbone comparison의	The “Roaring 20s” of visual recognition began with the introduction of Vision Transformers (ViTs), which quickly superseded ConvNets as	OpenAlex abstract

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					자연스러운 기준점이 된다.	the state-of-the-art image classification model.	
21	4	Expanding ERF while Maintaining Asymptotically Gaussian Distribution	Related Work; Discussion	무조건적 확대보다 안정적 ERF expansion이 중요하다는 최신 흐름	Recent ERF design work suggests that enlarging receptive fields while maintaining stable distributional behavior may be more important than simply maximizing kernel size.	Rather than merely employing extremely large ERF, it is more effective and efficient to expand the ERF while maintaining AGD of ERF by proper combination of smaller kernels.	Chrome/abstract-manual
22	5	Large Kernel Adapter for Enhanced Medical Image Classification	Related Work; Discussion	과도한 large-kernel expansion이 항상 최선이 아니라는 task-dependent 근거	Large-kernel adaptation studies indicate that receptive field expansion remains task-dependent, supporting our interpretation that moderate ERF expansion can outperform excessive enlargement.	Medical images exhibit extensive anatomical variation and low contrast, necessitating a large receptive field to capture critical features.	Chrome/abstract-manual

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23	6	Exploiting Gaussian based effective receptive fields for object detection	Discussion	ERF를 설계 변수로 다룰 수 있다는 관점 제시	ERF can be treated not only as an analysis target but also as a controllable design variable, opening room for adaptive ERF extensions in future sea fog classification work.	The effective receptive field (ERF) is a crucial concept in object detection, as it captures rich semantic information about the target, including its position and class.	OpenAlex abstract
24	7	Spatially-Adaptive Gradient Re-parameterization for 3D Large Kernel	Related Work; Discussion	large-kernel 설계에서도 spatial emphasis distribution이 중요함을 강조	Large-kernel performance depends not only on kernel size itself but also on how spatial emphasis is distributed, which aligns with our concern over where the model attends under fog.	Large kernel convolutions offer a scalable alternative to vision transformers for high-resolution 3D volumetric analysis, yet naively increasing kernel size often leads to optimization instability.	OpenAlex abstract
25	8	Expert-Like Reparameterization of Heterogeneous Pyramid Receptive Fields	Related Work; Methods	heterogeneous receptive field 조합의 타당성 제시	Heterogeneous receptive-field combinations provide a useful conceptual basis for our branch-style design that balances local and broader spatial cues.	Efficient convolutional neural network (CNN) architecture design has attracted growing research interests.	OpenAlex abstract
26	9	Numerical investigation of ERF and its	Related Work; Discussion	같은 kernel enlargement라도 architecture	The relationship between kernel	The receptive field (RF) plays a crucial role	Chrome/abstract-manual

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		relationship with kernels		에 따라 실제 ERF 변화가 다를 수 있음을 설명	size and actual ERF is not purely linear, which helps explain why the same enlargement can behave differently across backbones.	in convolutional neural networks (CNNs) because it determines the amount of input information that each neuron in a CNN can perceive, which directly affects the feature extraction ability.	
27	10	Multi-Scale Context Aggregation by Dilated Convolutions	Related Work	broader context aggregation이라는 receptive field 확장의 일반 목적 제시	Even when using dilation instead of large kernels, the underlying goal remains broader context aggregation, which is the same design motivation pursued in this study.	State-of-the-art models for semantic segmentation are based on adaptations of convolutional networks that had originally been designed for image classification.	OpenAlex abstract

Backbone

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28	17	Xception: Deep Learning with Depthwise Separable Convolutions	Related Work; Methods; Discussion	small-kernel depthwise backbone의 대표 사례	We include Xception as a representative depthwise-separable backbone because its spatial filtering relies heavily on small-kernel depthwise operations, making it suitable for testing	We present an interpretation of Inception modules in convolutional neural networks as being an intermediate step in-between regular convolution and the depthwise separable	OpenAlex abstract

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					local shortcut susceptibility.	convolution operation (a depthwise convolution followed by a pointwise convolution).	
29	18	MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications	Related Work	depthwise separable convolution 기반 경량 backbone 계보의 출발점	Since MobileNets established depthwise separable convolution as a standard building block for efficient vision models, receptive field design within this family becomes an important research question.	We present a class of efficient models called MobileNets for mobile and embedded vision applications.	OpenAlex abstract
30	19	MobileNetV2: Inverted Residuals and Linear Bottlenecks	Related Work	경량 backbone의 표현력 확장 계보	MobileNetV2 improved representational efficiency while retaining a small-kernel design, making it relevant when discussing how ERF redesign may help lightweight architectures.	In this paper we describe a new mobile architecture, MobileNetV2, that improves the state of the art performance of mobile models on multiple tasks and benchmarks as well as across a spectrum of different model sizes.	OpenAlex abstract
31	20	Searching for MobileNetV3	Related Work; Methods	실용적 small-kernel 경량 backbone 대표	MobileNetV3 serves as a practical small-kernel backbone for evaluating whether ERF redesign remains beneficial under efficiency-oriented architectures.	We present the next generation of MobileNets based on a combination of complementary search techniques as well as a novel architecture design.	OpenAlex abstract
32	21	EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks	Related Work; Discussion	scaling만으로는 부족하고 ERF 설계도 중요하다는 대비 근거	EfficientNet highlights the value of model scaling, but our problem setting suggests that scaling alone may not resolve shortcut reliance without receptive field redesign.	Convolutional Neural Networks (ConvNets) are commonly developed at a fixed resource budget, and then scaled up for better accuracy if more resources are available.	OpenAlex abstract

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33	22	EfficientNetV2: Smaller Models and Faster Training	Related Work; Methods	효율적 modern CNN backbone baseline	EfficientNetV2 provides an efficient modern CNN baseline, allowing us to test whether ERF redesign matters beyond standard scaling improvements.	This paper introduces EfficientNetV2, a new family of convolutional networks that have faster training speed and better parameter efficiency than previous models.	OpenAlex abstract
34	23	Deep Residual Learning for Image Recognition	Related Work	깊이 증가와 ERF 활용 문제를 구분하는 기준점	Residual learning made very deep networks practical, but increasing depth alone does not guarantee that a model will rely on the right scene-level cues.	Deeper neural networks are more difficult to train.	OpenAlex abstract
35	24	Swin Transformer: Hierarchical Vision Transformer using Shifted Windows	Related Work; Methods	broader context를 모델링하는 non-CNN baseline	We additionally compare against a hierarchical transformer baseline, since Swin explicitly integrates broader context through shifted windows.	This paper presents a new vision Transformer, called Swin Transformer, that capably serves as a general-purpose backbone for computer vision.	OpenAlex abstract
36	25	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	Related Work	global context modeling의 대조축	Vision Transformers directly model long-range context, providing a useful contrast to our goal of improving context usage within CNN backbones.	While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited.	OpenAlex abstract
37	26	ImageNet Classification with Deep Convolutional Neural Networks	Related Work	CNN 기반 image classification의 역사적 기준점	Since AlexNet established deep CNNs as the dominant paradigm for image classification, later questions about bias, context, and receptive field design naturally arise within this lineage.	We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes.	OpenAlex abstract

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38	32	Deformable Convolutional Networks	Related Work; Discussion	adaptive receptive field alternative	Deformable convolution offers an adaptive alternative to fixed kernel enlargement, and therefore serves as a useful reference point when discussing future adaptive ERF designs.	Convolutional neural networks (CNNs) are inherently limited to model geometric transformations due to the fixed geometric structures in their building modules.	OpenAlex abstract

해무 도메인

No.	기존번호	논문	넣을 파트	담당 논리	논문에 넣을 문장	원문 근거 문장	근거 출처
39	35	Spatio-Temporal Network for Sea Fog Forecasting	Related Work	CCTV sea fog literature가 forecasting으로 확장되었음을 보여줌	Prior CCTV-based sea fog research has extended toward spatio-temporal forecasting, but the effect of backbone receptive field design on shortcut behavior in single-frame classification remains underexplored.	Sea fog can seriously affect schedules and safety by reducing visibility during marine transportation.	OpenAlex abstract
40	36	RDCP: A Real Time Sea Fog Intensity and Visibility Estimation Algorithm	Introduction; Discussion	실시간 해무 강도/시정 추정이라는 응용 맥락	Real-time sea fog intensity and visibility estimation studies reinforce that sea fog recognition is operationally meaningful beyond an offline classification benchmark.	A number of accidents at sea are primarily caused by low visibility due to sea fog.	OpenAlex abstract

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41	38	Sea fog detection based on unsupervised domain adaptation	Introduction; Related Work; Discussion	annotation difficulty와 domain gap 문제의 직접 근거	Sea fog detection suffers from annotation difficulty and domain gap, supporting our decision to treat inter-port camera shift and boundary-label instability as first-order issues.	However, because few meteorological observations and buoys over the sea can be realized, obtaining visibility information to help the annotations is difficult.	Chrome/abstract-manual
42	39	Sea Fog Identification from GOCI Images using CNN Transfer Learning	Related Work; Methods	해무 도메인에서도 transfer learning이 자연스러운 접근임을 보강	Transfer-learning-based sea fog identification studies support our decision to initialize from generic visual pretraining before fine-tuning on the target fog dataset.	This study proposes an approaching method of identifying sea fog by using Geostationary Ocean Color Imager (GOCI) data through applying a Convolution Neural Network Transfer Learning (CNN-TL) model.	OpenAlex abstract
43	40	Multi-Satellite Image Matching for Daytime Sea Fog Detection	Related Work; Discussion	ground truth 구축과 label quality 중요성 보강	Recent sea fog detection research has highlighted the importance of careful ground-truth construction, which supports our emphasis on label auditing before drawing conclusions	Traditionally, sea fog detection technologies have relied primarily on in situ observations.	OpenAlex abstract

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					about ERF effects.		
44	41	A scSE-LinkNet Deep Learning Model for Daytime Sea Fog Detection	Related Work	attention-enhanced sea fog segmentation 흐름 제시	Attention-enhanced sea fog segmentation studies show that improving feature utilization matters in this domain, although our focus is on receptive field design rather than attention modules.	Sea fog is a precarious weather disaster affecting transportation on the sea.	OpenAlex abstract
45	42	Daytime Sea Fog Identification Based on ECA-TransUnet Model	Related Work	transformer/hybrid approaches가 broader context를 중시함을 보여줌	Hybrid transformer-based sea fog identification methods further suggest that broader context matters in this domain, even though our work seeks that benefit through CNN ERF redesign.	Sea fog is a weather hazard along the coast and over the ocean that seriously threatens maritime activities.	OpenAlex abstract
46	43	Short-Term Sea Fog Area Forecast: A New Data Set and Deep Learning Approach	Related Work; Discussion	해무 데이터셋 구축과 딥러닝 예측 literature 연결	Short-term sea fog nowcasting literature emphasizes both dataset construction and deep learning prediction, providing a broader context for future	Abstract Prompt and precise forecast of sea fog regions ensures maritime navigational safety.	OpenAlex abstract

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					extensions of our classification study.		
47	44	FogNet: A Multiscale 3D Convolutional Neural Network with Double-Branch Dense Block for End-to-End Fog Prediction from Satellite Data	Related Work; Discussion	multi-scale, multi-branch 설계가 fog prediction에 유효함을 제시	Fog prediction studies using multiscale and multi-branch architectures support the idea that combining multiple spatial ranges can be beneficial under fog conditions.	The reduction of visibility adversely affects land, marine, and air transportation.	OpenAlex abstract
48	45	Correlation context-driven method for sea fog detection	Introduction; Related Work; Discussion	sea fog detection에서 context가 중요하다는 가장 직접적인 도메인 근거	Sea fog detection itself has been argued to require correlation context beyond isolated local patterns, which is consistent with our emphasis on scene-level structural visibility cues.	Sea fog detection is a challenging and essential issue in satellite remote sensing.	OpenAlex abstract

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49	46	Machine learning analysis and nowcasting of marine fog visibility	Introduction; Discussion	visibility가 연속적 현상이라는 broader framing 보강	Marine fog visibility can be treated as a continuous phenomenon, suggesting that our normal-lowvis-seafog taxonomy approximates an underlying visibility continuum.	Introduction: This study presents the application of machine learning (ML) to evaluate marine fog visibility conditions and nowcasting of visibility based on the FATIMA (Fog and turbulence interactions in the marine atmosphere) campaign observations collected during July 2022 in the North Atlantic in the Grand Banks area and vicinity of Sable Island, northeast of Canada.	OpenAlex abstract
50	47	Data-to-data translation-based nowcasting of specific sea fog using geostationary weather satellite observation	Discussion	sea fog를 시간적으로 변화하는 현상으로 보는 future work 연결	Data-to-data sea fog nowcasting studies underscore that fog is a dynamic process, motivating future extensions of our work toward temporal modeling.	Dense sea fog events are responsible for many traffic accidents and fatalities.	Chrome/abstract-manual
51	48	Sea fog detection using	Related Work; Discussion	global context 활용이 해무 탐지에 유효하다는 보조 근거	Transformer-based sea fog detection	[Open abstract not found] Sea fog detection	Title-level fallback

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		transformer-based deep learning			results further suggest that long-range contextual reasoning is beneficial in this domain.	using transformer-based deep learning.	
52	55	Semantic Foggy Scene Understanding with Synthetic Data	Related Work; Discussion	foggy scene annotation difficulty, synthetic fog, dehazing 한계	Foggy-scene understanding studies show that real fog annotation is difficult and that synthetic fog plus supervision transfer can be more effective than relying on dehazing alone.	Due to the difficulty of collecting and annotating foggy images, we choose to generate synthetic fog on real images that depict clear-weather outdoor scenes, and then leverage these partially synthetic data for semantic foggy scene understanding.	Chrome/abstract-manual
53	new	Curriculum Model Adaptation with Synthetic and Real Data for Semantic Foggy Scene Understanding	Related Work; Discussion	light synthetic fog에서 dense real fog로의 staged adaptation 필요성	Curriculum adaptation studies suggest that foggy-scene recognition requires staged adaptation from synthetic light fog to dense real fog, underscoring how severe the domain gap can be in adverse-weather vision.	We propose a novel method, named Curriculum Model Adaptation, which gradually adapts a semantic segmentation model from light synthetic fog to dense real fog in multiple steps, using both labeled synthetic foggy data and unlabeled	Chrome/abstract-manual

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						real foggy data.	
54	new	A Multi-Feature Fusion Approach for Sea Fog Detection Under Complex Background	Introduction; Related Work; Discussion	complex background에서 global context, long-range dependency, multi-scale fusion 필요성	Recent sea fog studies under complex background explicitly emphasize long-range dependency and global context, supporting our claim that broader scene-level visibility structure is more informative than isolated local texture in ambiguous fog scenes.	For large-scale sea fog regions, the self-attention mechanism in transformer model captures the intrinsic features and identifies long-range dependencies.	Chrome/abstract-manual

시정/라벨 불확실성

No.	기존 번호	논문	넣을 파트	담당 논리	논문에 넣을 문장	원문 근거 문장	근거 출처
55	new	VISOR-NET: Visibility Estimation Based on Deep Ordinal Relative Learning under Discrete-Level Labels	Introduction; Dataset limitation; Discussion	visibility level의 ordinal structure와 continuous transition 근거	VISOR-NET treats visibility estimation as an ordinal problem rather than a purely categorical one, suggesting that the lowvis-seafog boundary may lie on a continuous visibility transition rather than a clean hard split.	This paper proposes a novel end-to-end pipeline that uses the ordinal information and relative relation of images for visibility estimation (VISOR-NET).	Chrome/abstract-manual
56	new	Visibility Estimation Based on Weakly Supervised	Introduction; Discussion	uneven fog distribution과 discrete label	Visibility estimation studies that model discrete label	Firstly, we transform the original single labels into discrete label	Chrome/abstract-manual

No.	기존 번호	논문	넣을 파트	담당 논리	논문에 넣을 문장	원문 근거 문장	근거 출처
		Learning under Discrete Label Distribution		distribution 관점 제시	distributions explicitly exploit uneven fog distribution within an image, aligning well with our observation that strong foreground edges may coexist with severe distant-structure degradation.	distributions and introduce discrete label distribution learning on top of the existing classification networks to learn the difference in visibility information among different regions of an image.	
57	50	Visibility Estimation Based on Weakly Supervised Learning	Related Work; Discussion	weak supervision을 통한 visibility labeling difficulty 보강	Weakly supervised visibility estimation highlights the difficulty of acquiring high-quality visibility labels, which supports our cautious treatment of ambiguous lowvis-seafog samples.	This paper proposes an end-to-end neural network model that fully utilizes the characteristic of uneven fog distribution to estimate visibility in fog images.	OpenAlex abstract
58	51	Highway Visibility Estimation in Foggy Weather via Multi-Scale Fusion Network	Related Work; Discussion	foggy visibility estimation에서 multi-scale fusion 중요성	Foggy-scene visibility estimation benefits from multi-scale fusion, reinforcing our claim that local edge and broader structural cues should be considered together.	Poor visibility has a significant impact on road safety and can even lead to traffic accidents.	OpenAlex abstract
59	52	Multi visual feature fusion based fog visibility estimation for expressway surveillance	Related Work	다중 시각 특징 융합이 visibility estimation에 유효함을 제시	Multi-feature visibility estimation studies suggest that no single local cue is sufficient,	[Open abstract not found] Multi visual feature fusion based fog visibility estimation for expressway	Title-level fallback

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		using deep learning network			supporting the need to aggregate multiple spatial cues in fog recognition.	surveillance using deep learning network.	
60	54	Deep Multi-head Regression Network for Pixel-wise Visibility Estimation from Hazy Images	Discussion	visibility가 spatial field처럼 분포할 수 있음을 보강	Pixel-wise visibility estimation implies that visibility can vary spatially across a scene, which supports our distinction between clear local edges and degraded distant structures.	Scene perception is essential for driving decision-making and traffic safety.	OpenAlex abstract
61	56	Surveillance Camera-Based Deep Learning Framework for Precipitation Type Classification	Related Work; Discussion	surveillance camera 기반 weather classification의 camera-specific bias 문제	Surveillance-camera weather classification studies indicate that camera viewpoint, noise, and lighting can strongly affect generalization, which parallels our inter-port domain-bias concerns.	Urban surveillance cameras offer a valuable resource for high spatiotemporal resolution observations of near surface precipitation type (SPT), with significant implications for sectors such as transportation, agriculture, and meteorology.	Chrome/abstract-manual

학습/평가

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62	57	ImageNet Large Scale Visual Recognition Challenge	Methods	사전학습 정당화	Because the sea fog dataset is relatively limited in scale, we adopt ImageNet-based pretraining as a standard	The ImageNet Large Scale Visual Recognition Challenge is a benchmark in object category classification and detection on	Chrome/abstract-manual

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					initialization strategy.	hundreds of object categories and millions of images.	
63	58	ImageNet pre-training and two-step transfer learning	Methods	2-stage transfer setting 정당화	Following recent two-step transfer learning practice, we first pretrain on a generic image dataset and then fine-tune on the port CCTV sea fog dataset.	Chromosome image classification typically relies on ImageNet pre-training, yet the potential of leveraging intermediate domains from related staining techniques remains largely underexplored.	OpenAlex abstract
64	59	How transferable are features in deep neural networks?	Related Work; Discussion	layer-wise transferability 차이를 ERF redesign 해석과 연결	Layer-wise transferability studies suggest that the effect of ERF redesign may differ depending on where in the backbone the relevant features are formed.	Many deep neural networks trained on natural images exhibit a curious phenomenon in common: on the first layer they learn features similar to Gabor filters and color blobs.	OpenAlex abstract
65	60	Decoupled Weight Decay Regularization	Methods	AdamW optimizer 선택 정당화	We optimize all models with AdamW, which decouples weight decay from the adaptive update and is widely used in modern vision training.	L$_2$ regularization and weight decay regularization are equivalent for standard stochastic gradient descent (when rescaled by the learning rate), but as we demonstrate this is \emph{not} the case for adaptive gradient algorithms, such as Adam.	OpenAlex abstract

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66	61	Focal Loss for Dense Object Detection	Discussion; Future Work	class imbalance 대응 loss 설계의 future-work 근거	Since lowvis-seafog confusion can be viewed through an imbalance lens, imbalance-aware objectives such as focal-type losses remain a plausible future extension.	The highest accuracy object detectors to date are based on a two-stage approach popularized by R-CNN, where a classifier is applied to a sparse set of candidate object locations.	OpenAlex abstract
67	63	YOLOv8	Discussion	실용형 비전 시스템의 효율성 지향 경향을 보조적으로 언급	Practical vision systems increasingly prioritize efficient inference, and a deployable sea fog classifier may eventually need to balance recognition quality with operational efficiency.	YOLO has become a central real-time object detection system for robotics, driverless cars, and video monitoring applications.	OpenAlex abstract
68	64	Enhanced YOLOv8 Based Drone Detection for Improving Operational Stability	Related Work; Discussion	저자 기존 실용형 vision refinement 경험과의 연결	Our previous operational-stability-oriented vision work provides a practical backdrop for framing sea fog classification as a deployable maritime perception component.	[Open abstract not found] Enhanced YOLOv8 Based Drone Detection for Improving Operational Stability.	Title-level fallback
69	65	Feature Pyramid Networks for Object Detection	Related Work; Discussion	multi-scale representation의 일반적 중요성 보조 근거	Multi-scale representation has long been valuable in visual recognition, supporting our view that fog classification	Feature pyramids are a basic component in recognition systems for detecting objects at different scales.	OpenAlex abstract

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					should not depend on a single local scale.		
70	66/67	A systematic analysis of performance measures for classification tasks; Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation	Methods	macro-F1, recall, 다면 평가의 직접 정당화	We prioritize macro-F1, class-wise recall, and confusion analysis rather than relying solely on overall accuracy, because different metrics reflect different error characteristics in imbalanced classification problems.	Commonly used evaluation measures including Recall, Precision, F-Measure and Rand Accuracy are biased and should not be used without clear understanding of the biases.	Chrome/abstract-manual

사용 우선순위

Core 20

- 1, 2, 3, 4, 5, 9, 10, 11, 13, 18, 19, 28, 31, 33, 41, 43, 48, 55, 56, 70

Strong support 30

- 6, 7, 8, 12, 14, 15, 17, 20, 21, 22, 23, 24, 25, 26, 29, 30, 32, 35, 36, 37, 39, 42, 44, 45, 47, 49, 52, 53, 58, 62

Context / methods / future-work support 20

- 16, 27, 34, 38, 40, 46, 50, 51, 54, 57, 59, 60, 61, 63, 64, 65, 66, 67, 68, 69

집필 메모

- Introduction에서는 문제 중요성 -> shortcut/local bias -> ERF necessity -> sea fog context -> label ambiguity 순서를 유지한다.
- Related Work는 `ERF and large-kernel design`, `shortcut/local evidence bias`, `sea fog and visibility understanding`의 3축으로 정리한다.
- Methods에서는 backbone provenance, pretraining, optimizer, evaluation metric, Grad-CAM/perturbation 근거를 압축해서 쓴다.
- Discussion에서는 `ERF expansion이 shortcut을 줄였는가`와 `scene-level visibility cue가 무엇인가`를 문헌과 실험을 함께 엮어 닫는다.